



Dr. N.G.P. Institute
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EDUCATION

B.Tech, Artificial Intelligence and Data Science

Dr. N.G.P. Institute of Technology
2024 – 2028

CGPA: [x.xx]

TECHNICAL SKILLS

Languages

Python, JavaScript

Backend and Data

FastAPI, Node.js, Express, REST APIs, JWT auth, Pydantic, MongoDB, Beanie

Frontend

React

ML and CV

scikit-learn, XGBoost, YOLOv8, ByteTrack, MediaPipe, TensorFlow Lite, 1D-CNN, signal processing, causal uplift modeling

Tools

Git, GitHub Actions, aiogram (Telegram bots), Canva

CERTIFICATION

NPTEL Elite Certificate: Human-Computer Interaction

IIT Madras, 90% score

LANGUAGES

English, Telugu, Tamil

DHANVANTH

AI & Data Science Student | Full-Stack and Machine Learning Developer

PROFILE

Pre-final year B.Tech AI & Data Science student who builds end-to-end systems, from signal-processing and computer-vision models to REST APIs and React front ends. Published researcher (ICACIT 2025), hackathon participant and Joint Secretary of the Department of AI & Data Science association (ALZENIXX). Seeking a Software Engineering or ML/AI internship.

PROJECTS

ECG Arrhythmia Classifier [GitHub](#)

Python · scikit-learn · XGBoost

- From-scratch ML pipeline (no pretrained models) classifying **17 cardiac arrhythmia types** from single-lead ECG at **91.3% cross-validated accuracy**.
- Engineered 41 features per recording (HRV, spectral, wavelet) over 1,000 signals; Random Forest / SVM / XGBoost ensemble. Built at a hackathon at VIT Vellore.

Employee Skill & Project Allocation System [GitHub](#)

FastAPI · MongoDB · Beanie · JWT

- Role-based full-stack platform with a **skill-matching engine that ranks best-fit candidates** for each project.
- Clean layered backend (routes, schemas, services, models) with versioned **/api/v1** endpoints, async FastAPI and Pydantic validation.

ElevenIQ: Football Analytics and League Center [GitHub](#)

YOLOv8 · FastAPI · React · MongoDB

- Match analysis: player detection and tracking (YOLOv8 + ByteTrack), pitch homography, heatmaps, pass networks, sprint/fatigue metrics and an xG model.
- League Center with live standings, fixtures and results for the top-5 leagues and Champions League, auto-refreshing every 60 seconds.

Real-Time Neurological Tremor Detection

MediaPipe · TFLite · 1D-CNN

- Edge-AI camera system detecting tremors in real time with MediaPipe landmarks and a lightweight 1D-CNN on TensorFlow Lite; prepared IEEE-format base papers.

Asteroid Impact Detection System

ICACIT 2025

- Published a paper at ICACIT 2025** on an asteroid impact detection system.

LEADERSHIP AND ACTIVITIES

Joint Secretary, Dept. of AI & Data Science (ALZENIXX)

Dr. N.G.P. Institute of Technology

- Designed the **ALZENIXX department handbook** and the **INSIGHTX magazine** in Canva, with unique editorial layouts per page and real event content.

Class LeetCode Tracker [GitHub](#)

GitHub · 338+ commits

- Maintain a GitHub-based tracker of solved problems for a **class of 41 students** to keep DSA practice consistent.

ACHIEVEMENTS

- Published** at ICACIT 2025 (Asteroid Impact Detection System).
- 36-hour hackathon, NIT Trichy:** built "Counterfactual Classroom", a causal uplift model on the OULAD dataset.
- Hackathon, VIT Vellore:** ECG arrhythmia classification at 91.3% cross-validated accuracy.